labTrain-5Gcore-platform

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Course Information

- **Course Title:** 5Gcore Platform
- **Level:** Undergraduate
- **Duration:** 6-8 Weeks
- **Format:** Reading + presentation + Laboratory

Course Description

This course takes learners from zero telecom-specific knowledge to basically understanding and operating a 5G Core (5GC) network, This course for the one who have knowledge of networking layers, sockets, virtualization, and OS process/service models. This course maps directly from familiar CS concepts (client-server APIs, microservices, state machines) onto 5GC's Service-Based Architecture (SBA), where each Network Function exposes stateless HTTP/2 REST APIs and registers with a discovery service (NRF). The course is structured around 3GPP TS 23.501 (System Architecture) and TS 23.502 (Procedures), reinforced through weekly readings, instructor/peer presentations, and progressively harder labs using free5GC, Open5GS, and StormSIM.

The final project is a working lab environment where a simulated RAN and UE successfully executes UE registration and PDU session establishment against an open-source 5GC.

Learning Outcomes

By the end of this course, students will be able to:

- Explain the 5GC Service-Based Architecture (SBA) and differentiate it from legacy EPC/4G core design.
- Identify and describe the role of each core Network Function: AMF, SMF, UPF, PCF, UDM, NRF, NSSF.
- Trace key call flows including UE registration, PDU session establishment, and mobility/handover procedures across N1, N2, N3, N4, N6 interfaces.
- Deploy a working 5G Core using open-source tools and connect it to a simulated RAN.
- Capture and analyze NGAP, PFCP, and SBI (HTTP/2) protocol traffic using packet analysis tools.
- Configure basic network slicing (NSI/NSSI) and articulate its role in multi-tenant 5G networks (**OPTIONAL**).
- Final target: Implement a simple RAN-UE test setup that successfully performs registration and PDU session establishment against an open-source 5GC, and produce a documented protocol-level report of the exchange.

Textbook, Reference

- 3GPP TS 23.501 - System Architecture for 5G System
- 3GPP TS 23.502 – Procedures for the 5G System
- Signaling and Communication in 5G Mobile Networks (relevant chapters on SBA)
- Free5GC documentation

Course Structure

The course is organized into 5 stages, each combining reading (spec/textbook study), presentation (concept synthesis and peer teaching), and laboratory (hands-on deployment) components, following the principle that each layer of 5GC understanding builds on the previous one.

- Stage 1 5G System Architecture Foundations
- Stage 2 Network Functions Deep Dive (SBA)
- Stage 3 Protocols & Interfaces (N1–N6, NAS, NGAP, PFCP)
- Stage 4 Registration + PDU Session Establishment Procedure Deep Dive
- Stage 5 Capstone Integration: RAN-UE $\leftrightarrow$ 5GC Testing

Stage 1: 5G System Architecture Foundations

Reference: Reading: TS 23.501-chapter 4 (System Architecture), overview chapters on SA vs NSA deployment models, and the shift from EPC (4G) to SBA (5G).

Topics covered:

- Why 5GC abandons monolithic EPC for microservices-style SBA
- Control Plane vs User Plane separation (CUPS) rationale
- SA (Standalone) vs NSA (Non-Standalone) deployment models
- High-level NF list and system reference diagram

Stage 2: Network Functions Deep Dive (SBA)

Ref: TS 23.501-chapter 6 (Network Functions), documentation on AMF, SMF, UPF, PCF, UDM, NRF, NSSF, AUSF roles and their HTTP/2-based Service-Based Interfaces (SBI).

Topics covered:

- Access and Mobility Management Function (AMF) - connection/mobility state machine
- Session Management Function (SMF) - PDU session
- User Plane Function (UPF) - packet forwarding, QoS enforcement
- NRF as service discovery registry; PCF for policy; UDM for subscriber data
- Cloud-native deployment: NFs as Containerized Network Functions (CNFs) on Kubernetes

Lab:

- deploy free5GC or Open5GS
- (OPTIONAL) Install Docker/Kubernetes locally; deploy free5GC or Open5GS in a single-node cluster and verify all NF pods/services are running and registered with NRF.

Stage 3: Protocols and Interfaces

Ref: TS 23.502-chapter 4 (interfaces), NGAP protocol basics, NAS message structure, PFCP (N4) specification summary.

Topics covered:

- N1 (UE–AMF NAS signaling), N2 (gNB–AMF via NGAP), N3 (gNB–UPF user plane tunnel), N4 (SMF–UPF via PFCP), N6 (UPF–Data Network)
- HTTP/2 + JSON-based Service-Based Interfaces between NFs
- Introduction to Wireshark dissectors for NGAP, NAS-5GS, and PFCP

Lab: Capture idle-state SBI traffic between deployed NFs (from Stage 2) using Wireshark/tcpdump inside the cluster; identify NRF registration messages and PFCP heartbeat.

Stage 4: Registration Procedure Deep Dive

Ref: TS 23.502-chapter 4.2 vs 4.3 (Registration procedures + PDU Session procedures) — full call flow from RRC connection through NAS Registration Request to AMF, authentication (AUSF), and Registration Accept.

Topics covered:

- Initial Registration, Mobility Registration, Periodic Registration
- 5G-AKA authentication exchange sequence
- Security context establishment (NAS security mode command)
- PDU Session Establishment Request/Accept flow
- SMF-UPF N4 session setup via PFCP (Session Establishment/Modification Request)
- QoS flow mapping and default bearer equivalent in 5G
- IP address allocation for the UE's Data Network connection (N6)

Lab: Deploy StormSIM as a simulated gNB+UE; configure it to point at the Stage 2 5GC deployment; trigger UE registration + PDU Session Establishment request + verify UPF forwards user plane traffic to the internet/data network.

Stage 5: Final project - RAN/UE implementation

Lab (project): Build a complete test environment where a simulated RAN-UE performs full Registration + PDU Session Establishment against a deployed open-source 5GC (free5GC or Open5GS). Learners must produce: a working demo, a full Wireshark capture with message-by-message annotation, and a written report mapping observed behavior back to TS 23.502 procedures.